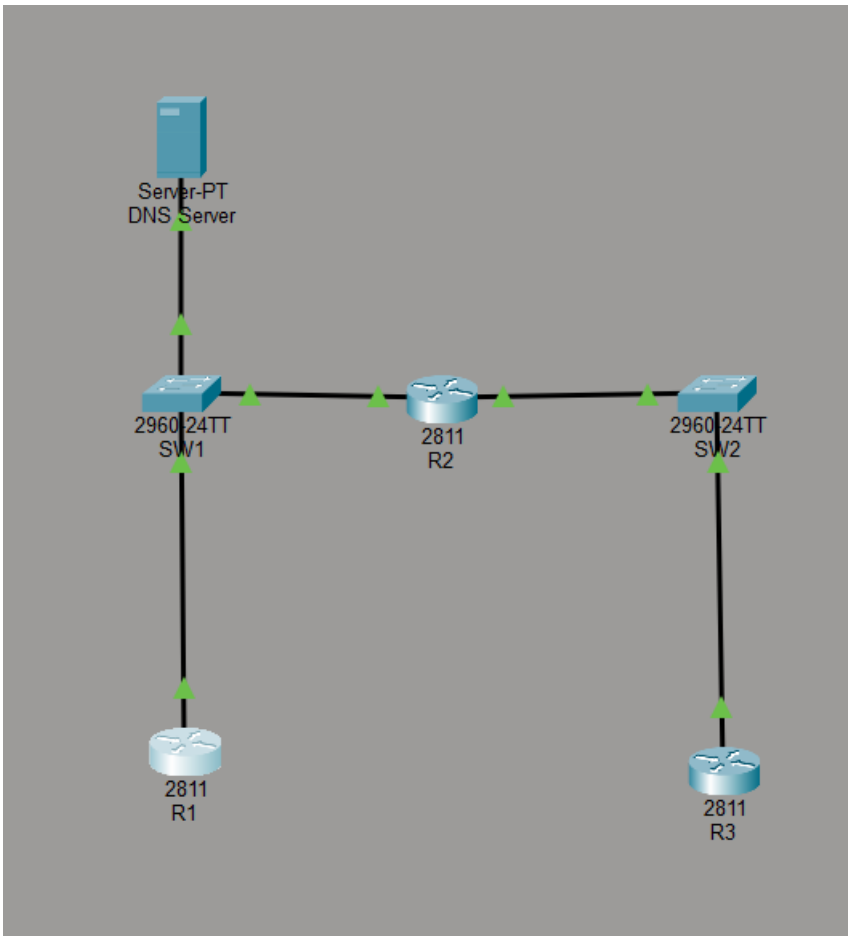
Network Topology Before Troubleshooting



I used my completed Week 3 network topology for this practical, the network contains three routers, two switches and one DNS server. I first confirmed that the network was working before introducing a fault.

Initial Connectivity Failure (Task 22)

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.30.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#
```

Copy Paste

I introduced a simulated network fault and tested connectivity from R1 to the DNS server using the ping command, the test failed, showing that there was a communication problem that needed troubleshooting.

Physical Layer Check (Task 23)

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.30.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#
```

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I checked the physical connections and the interface status on R1, the FastEthernet0/0 interface was up/up, and the cables were connected correctly, this showed that there was no obvious physical connection problem.

Data Link Layer Check (Task 23)

```
Switch>enable
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.85e1.a801    DYNAMIC     Fa0/1
   1    00e0.f920.5e01    DYNAMIC     Fa0/2
Switch#
```

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I checked the MAC address table on SW1 to see whether the switch was learning device addresses and the results helped me verify that the switch was forwarding traffic at the Data Link Layer.

Network Layer Check (Task 23)

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.30.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.30.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.30.0/24 is directly connected, FastEthernet0/0
L       10.10.30.1/32 is directly connected, FastEthernet0/0

R1#
```

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I checked the IP configuration and routing table on R1, and I discovered that FastEthernet0/0 was configured with 10.10.30.1 instead of 10.10.10.1, this placed R1 in the wrong subnet and caused the connectivity failure.

Correcting the Network Fault

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

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After identifying the incorrect IP address on R1, I changed FastEthernet0/0 back to 10.10.10.1 with the correct subnet mask, I then checked the interface status to confirm that the configuration was correct.

Test the Repaired Network (Task 25)

```
R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/2 ms

R1#
```

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After correcting the IP address on R1, I tested the network connection again using the ping command and both the DNS server and R2 responded successfully, this confirmed that the network fault had been resolved and communication was restored.

During this practical, I used the Cisco troubleshooting methodology to identify and fix a simulated network problem, I first tested the connection between R1 and the DNS server using the ping command, the test failed, which showed that there was a network problem. I then checked the Physical Layer by checking the cables and interface status. After that, I checked the Data Link Layer using the switch MAC address table, at the Network Layer, I discovered that R1 had the incorrect IP address of 10.10.30.1 instead of 10.10.10.1, I corrected the IP address and tested the connection again, the ping tests were successful, confirming that the problem was fixed.